Allwins Kannan Rajaiah

AI Solutions Architect

Agentic AI on AWS · AI-Driven Application Modernization · Data Platforms · Development & QA Team Leadership

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PROFESSIONAL SUMMARY

- AI Solutions Architect with 20+ years of experience in delivering enterprise applications, cloud-native platforms, and AI-enabled solutions, translating business challenges into scalable, secure, and production-ready architectures.
- Design end-to-end AI and cloud architectures on AWS, integrating Generative AI, Agentic AI, Java, Spring Boot, Angular, React, microservices, REST APIs, and enterprise systems to deliver business-aligned solutions.
- Build agentic AI solutions using Amazon Bedrock AgentCore, Strands Agents, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), and human-in-the-loop controls, incorporating secure tool integration, observability, and evaluation.
- Lead cloud-native application architecture and AI-enabled data engineering, combining enterprise application modernization with Databricks, Auto Loader, Lakeflow, PySpark, SQL, dbt, and Unity Catalog to deliver scalable data platforms, governed data products, automated reconciliation, and AI-powered exception investigation.
- Drive technology strategy and engineering excellence through governance, solution design reviews, technology roadmaps, security standards, and cross-functional leadership of development and QA teams, using AI to improve engineering productivity and operational outcomes.

CORE TECHNOLOGY

Architecture & Leadership: Solution Architecture, Cloud Architecture, Architecture Governance, Design Reviews, Enterprise Modernization, Team Leadership, Estimation & Release Readiness.

AI & GenAI: Generative AI, Agentic AI, Amazon Bedrock, AgentCore, Strands Agents, MCP, RAG, Bedrock Knowledge Bases, LLM Applications, AI Governance, AI Evaluation, Langfuse.

Cloud & Platform: AWS, Docker, Kubernetes, Terraform, Serverless, Cloud Security, Scalability & Cost Optimization.

Data Engineering: Databricks, Lakeflow, Auto Loader, Databricks Workflows, Unity Catalog, PySpark, dbt, SQL, Data Pipelines.

Application & Integration: Java, Spring Boot, Hibernate, Angular, Microservices, REST APIs, Redis, OAuth, JWT, Node.js, HTML, CSS, JavaScript, TypeScript.

DevOps & Observability: CI/CD, Git, Jenkins, AWS CodePipeline, Prometheus, Splunk, Datadog, OpenTelemetry.

Databases: Oracle, MySQL, DB2, Amazon RDS.

AI-Assisted Development: Claude Code, GitHub Copilot, Replit, Cursor.

PROFESSIONAL EXPERIENCE

Director of Technology | AI Solutions Architect

Tekizma, India · Sep 2024 – Present

- Lead enterprise technology direction by defining solution architecture, target-state roadmaps, and design standards for application, data, cloud, and AI platforms, while overseeing architecture governance across development and QA teams.

Enterprise Application Modernization

- Led the modernization of a legacy Spring/JSP enterprise application into a cloud-native platform built on Spring Boot, Angular, REST APIs, Hibernate, MySQL, and Redis, delivered with Docker containerization, Terraform-based infrastructure provisioning, AWS CodePipeline CI/CD, and end-to-end observability using OpenTelemetry, Datadog, and Splunk.
- Defined the target AWS architecture across EC2, ECS/Fargate, and Lambda, with VPC networking, ALB/NLB, and CloudFront for delivery, secured through IAM, KMS, Secrets Manager, Cognito, and WAF, and monitored centrally via CloudWatch.
- Developed AI-powered application capabilities using Agentic AI and Amazon Bedrock AgentCore, enabling intelligent, task-oriented workflows integrated with enterprise application services and business processes.
- Designed secure AI agent integration using the Model Context Protocol (MCP), enabling AI agents to interact with application tools, APIs, and enterprise services through controlled, standardized interfaces.
- Implemented RAG-based AI workflows with an enterprise knowledge base, leveraging enterprise knowledge retrieval and Claude Code to support intelligent information access, contextual responses, and AI-assisted application functionality.
- Established AI-Driven Development Life Cycle (AIDLC) practices across requirements analysis, solution design, implementation, testing, and validation, incorporating structured human review and quality gates to improve delivery consistency, maintainability, and engineering governance.

Key achievement: Accelerated engineering productivity using Claude Code, Agentic AI, MCP, and RAG to streamline code development, enable intelligent workflows, and improve contextual access to enterprise knowledge.

Technologies: Spring Boot, Angular, Hibernate, MySQL, Redis, Docker, Terraform, AWS CodePipeline, OpenTelemetry, Datadog, Splunk, Amazon Bedrock, AgentCore (Runtime, Gateway, Identity), Strands Agents, MCP, RAG, Bedrock Knowledge Bases, Claude Code.

Databricks Data Engineering & Reconciliation

- Built a scalable Databricks data engineering platform supporting incremental data ingestion through Auto Loader and Lakeflow, data modeling using PySpark, SQL, and dbt, and workflow orchestration through Databricks Workflows.
- Established data governance and reusable data-product capabilities using Unity Catalog, enabling governed data access and natural-language interaction with enterprise data, while leveraging GitHub Copilot to accelerate engineering productivity.
- Designed the end-to-end data reconciliation architecture, defining source-to-target mappings, data quality and validation rules, exception-handling mechanisms, and reconciliation controls across ingestion and transformation layers; implemented continuous delivery workflows using GitHub Actions.
- Developed an AI reconciliation agent to investigate Databricks pipeline exceptions, correlate failures with source-to-target mappings and operational runbooks, and recommend corrective actions for analyst review and approval. Integrated Langfuse for execution tracing and evaluation.

Key achievement: Improved reconciliation accuracy by 50% through automated validation, standardized exception handling, and AI-assisted exception investigation.

Technologies: Databricks, Auto Loader, Delta Live Tables, Databricks Workflows, Unity Catalog, PySpark, SQL, dbt, Langfuse, GitHub Actions, GitHub Copilot.

Solutions Architect

Advanced Software Design, Inc., USA · Sep 2013 – Aug 2024

- Delivered multi-tier enterprise applications using Spring Boot, Angular, microservices, REST APIs, and AWS, with secure, high-performance APIs covering authentication, authorization, and operational readiness.
- Designed AWS deployments on EC2, ECS, Lambda, and Amazon RDS, using VPC, ELB, IAM, S3, and CloudFront for networking, load balancing, access control, storage, and content delivery.
- Defined service boundaries, API contracts, and integration patterns for complex enterprise workflows.
- Directed application modernization, assessing legacy components and defining incremental migration strategies.

- Introduced reusable design patterns and technical components, improving consistency and development efficiency.
- Collaborated with development, QA, business analysis, and delivery teams on estimation, release readiness, and resolution of complex production issues.

Key achievement: Delivered the official Health Plan Marketplace for New York State of Health as Solutions Architect, supporting 5 million enrollments in 2020 at a peak load of concurrent users.

Technologies: Spring Boot, Angular, REST APIs, Microservices, AWS (EC2, ECS, Lambda, S3, RDS, VPC, ELB, IAM, CloudFront), Hibernate, Oracle, DB2, Docker, Kubernetes, Jenkins, Git.

Technical Manager | Mobile Solutions Architecture

Marlabs Software Pvt. Ltd. · Sep 2010 – Sep 2013

- Defined cross-platform mobile solution architecture for iOS, Android, and PhoneGap, covering security, authentication, REST APIs, and logging, and produced application designs, SDLC plans, and delivery roadmaps.
- Delivered eFleetPro enterprise integration solutions connecting construction-equipment manufacturers, dealers, and rental fleets with ERP systems, punch-out integrations, and payment gateways.
- Established signed mobile-to-enterprise integration patterns for secure communication with backend services and external systems.
- Led the Mobile Centre of Excellence, setting platform standards and managing team formation, hiring, and delivery across client and internal initiatives.
- Presented cross-platform capabilities and technology solutions to customers and leadership, supporting client engagement and revenue growth.

Technologies: iOS, Android, PhoneGap, REST APIs, Java, Mobile Architecture, Enterprise Integration, ERP Integration, Payment Gateway Integration, Security & Authentication.

ADDITIONAL EXPERIENCE

Tech Lead, Mapview Geographical Solutions, Sep 2009 – Aug 2010.

Tech Lead / Senior J2EE, Star4ce Technologies, Inc., Nov 2007 – Jun 2009.

Team Lead, Amtex Systems, Inc., Apr 2007 – Oct 2007.

Team Lead, 3i-Infotech, Apr 2004 – Feb 2007.

Java Developer, Asic Infotech Pvt. Ltd., May 2003 – Feb 2004.

EDUCATION

B.E., Computer Science & Engineering, Bharathiar University, India.

CERTIFICATION

AWS Certified AI Practitioner (AIF-C01). Issued September 2026. Credential ID cd4be8c7c3ea48aca09bc7a181355061. Verification: https://www.credly.com/earner/earned/badge/496bc744-8636-4032-9eb0-975867c4f0d5.